Gebze Technical University

Department of Mechanical Engineering

ME 331 System Dynamics and Vibration

Harmonic Force Response

Single-degree-of-freedom systems

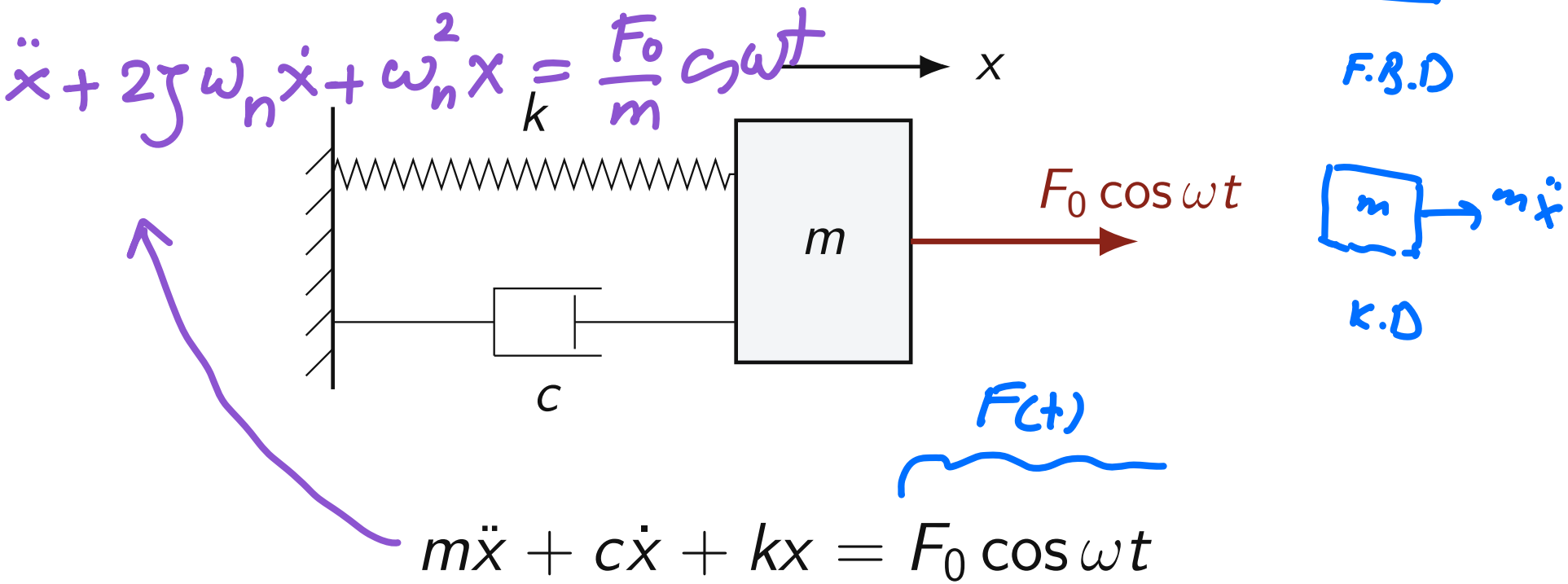
Saeed Lotfan

Today's topics

1. Harmonic excitation
2. Undamped forced response
3. Viscously damped response
4. Frequency-response curves
5. Resonance and damping
6. Worked application and checks

5.1 Harmonic excitation

An applied force drives the mass

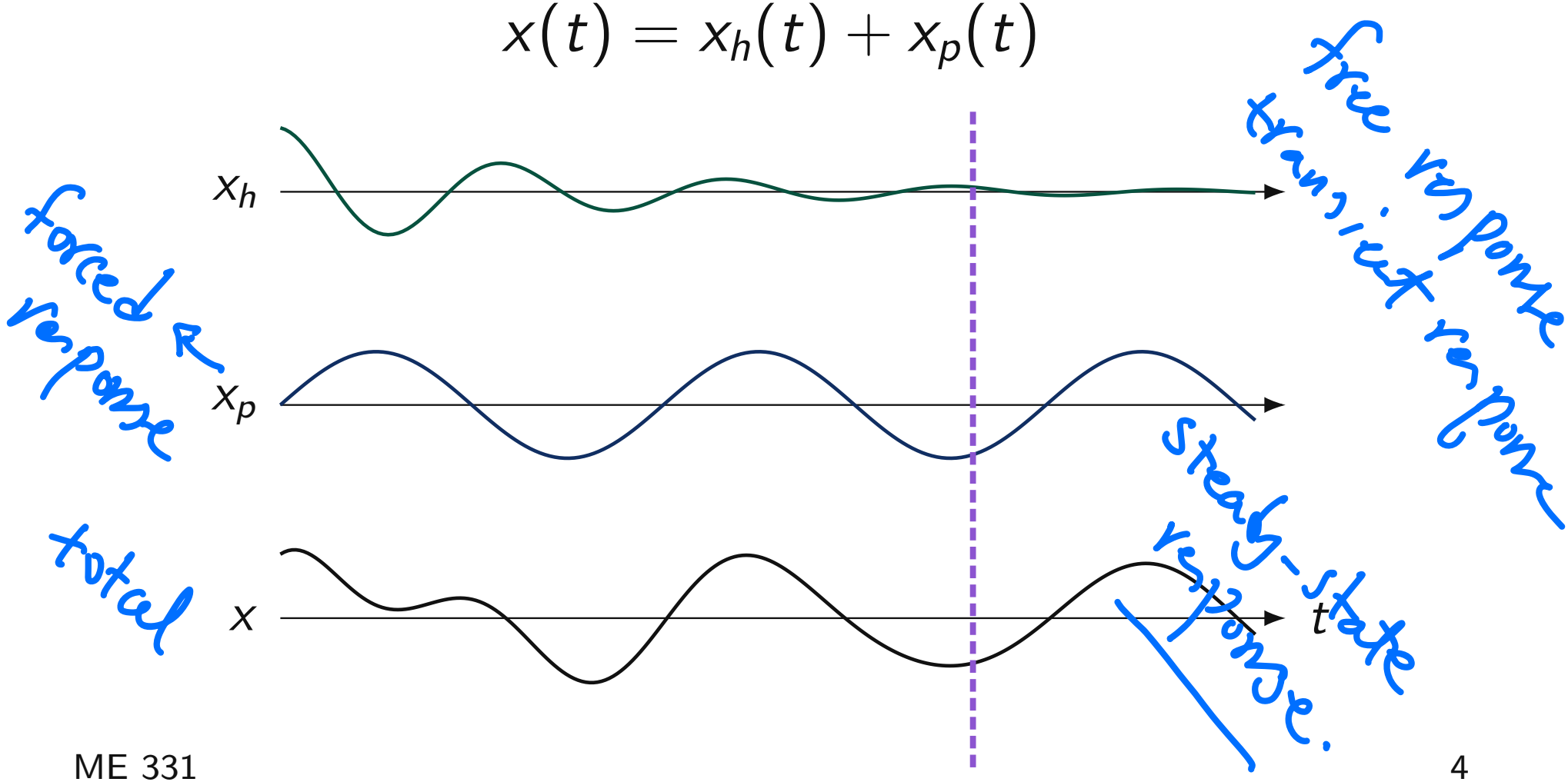


Which frequency is imposed, and which belongs to the system?

5.1 Harmonic excitation

Total response = free part + forced part

$$x(t) = x_h(t) + x_p(t)$$



5.2 Undamped forced response

Apply the initial conditions to the total motion

$$\ddot{x} + \omega_n^2 x = \frac{F_0}{m} \cos \omega t$$

$$x_p = \frac{F_0}{k - m\omega^2} \cos \omega t, \qquad \omega \neq \omega_n$$

Start from rest: $x(0) = 0$, $\dot{x}(0) = 0$.

What changes when $\omega > \omega_n$?

Whiteboard notes

$$m\ddot{x} + kx = F_0 \cos\omega t$$

$$x_h = A\cos(\omega_n t - \phi) = B_1 \sin\omega_n t + B_2 \cos\omega_n t; \quad \omega_n = \sqrt{\frac{k}{m}}$$

$$x_p = X\cos\omega t \rightarrow \ddot{x}_p = -X\omega^2 \cos\omega t$$

$$-mX\omega^2 \cos\omega t + kX\cos\omega t = F_0 \cos\omega t$$

$$X = \frac{F_0}{k - m\omega^2} \rightarrow x_p(t) = \frac{F_0}{k - m\omega^2}\cos\omega t$$

Whiteboard notes

$$x_{\text{total}}(t) = B_1 \sin \omega_n t + B_2 \cos \omega_n t + \frac{F_0}{k - m\omega^2} \cos \omega t$$

$$\begin{matrix} x(0) = 0 \\ \dot{x}(0) = 0 \end{matrix} \rightarrow \quad x(0) = B_2 + \frac{F_0}{k - m\omega^2} = 0 \qquad B_2 = \frac{-F_0}{k - m\omega^2}$$

$$\dot{x}(0) = B_1 \omega_n = 0 \rightarrow B_1 = 0$$

$$x(t) = \frac{F_0}{k - m\omega^2} \left(\cos \omega t - \cos \omega_n t \right)$$

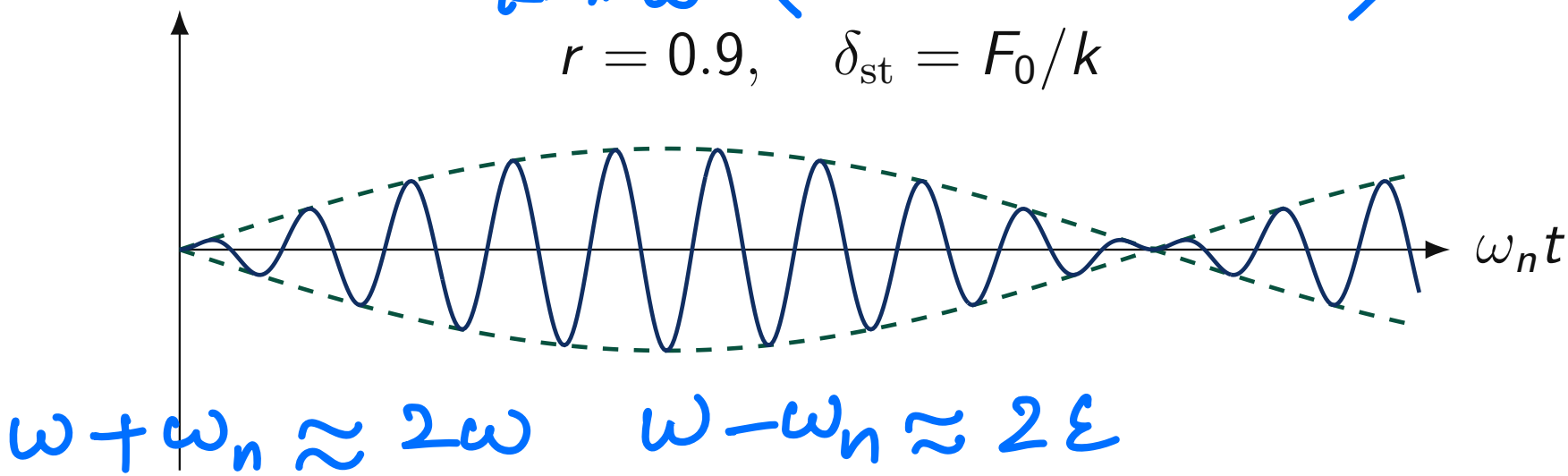
Static force $\rightarrow$ $\omega = 0 \rightarrow F(t) = F_0$

5.2 Undamped forced response

Nearby frequencies produce beating

$\omega \approx \omega_n$

$$x(t) = \frac{F_0}{k - m\omega^2}\left(\cos\omega t - \cos\omega_n t\right)$$



$\omega + \omega_n \approx 2\omega \qquad \omega - \omega_n \approx 2\varepsilon$

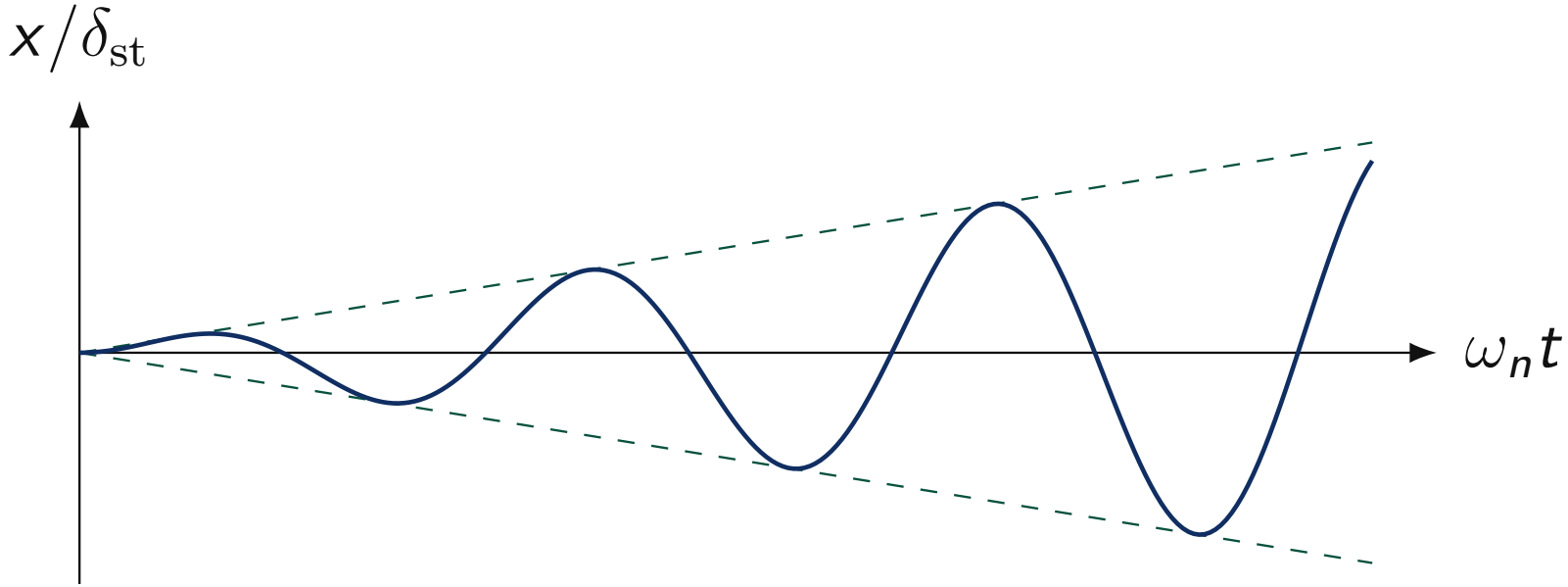
$$\cos\omega t - \cos\omega_n t = -2\sin\frac{(\omega+\omega_n)t}{2}\sin\frac{(\omega-\omega_n)t}{2}$$

5.2 Undamped forced response

$\omega = \omega_n$

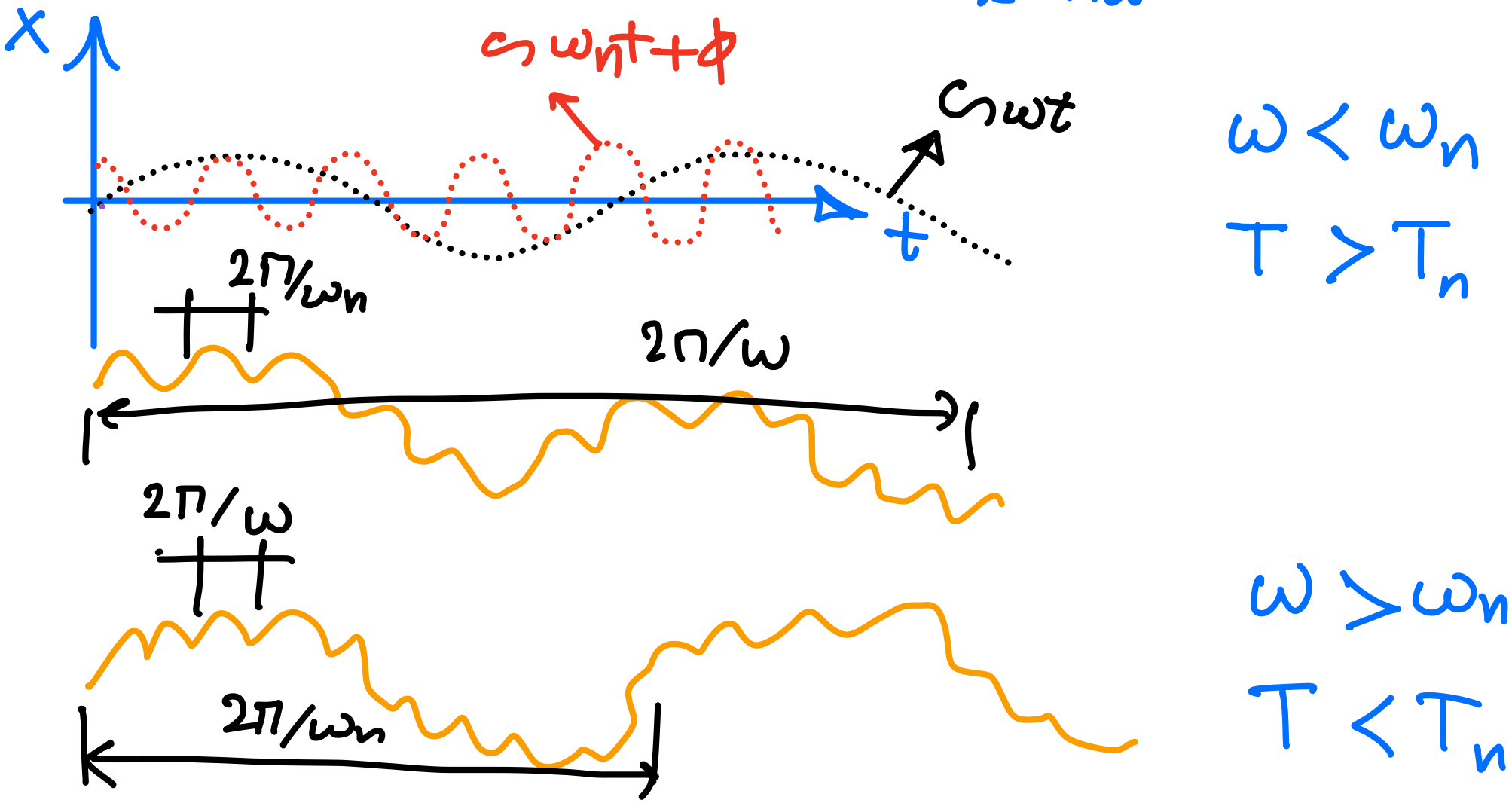
Exact resonance: growth with time

$$\omega = \omega_n, \quad x_0 = v_0 = 0: \qquad x(t) = \frac{F_0}{2m\omega_n} t \sin \omega_n t$$



Whiteboard notes

$$x(t) = B_1 \sin \omega_n t + B_2 \cos \omega_n t + \frac{F_0}{k - m\omega^2} \cos \omega t$$



$\omega < \omega_n$

$T > T_n$

$\omega > \omega_n$

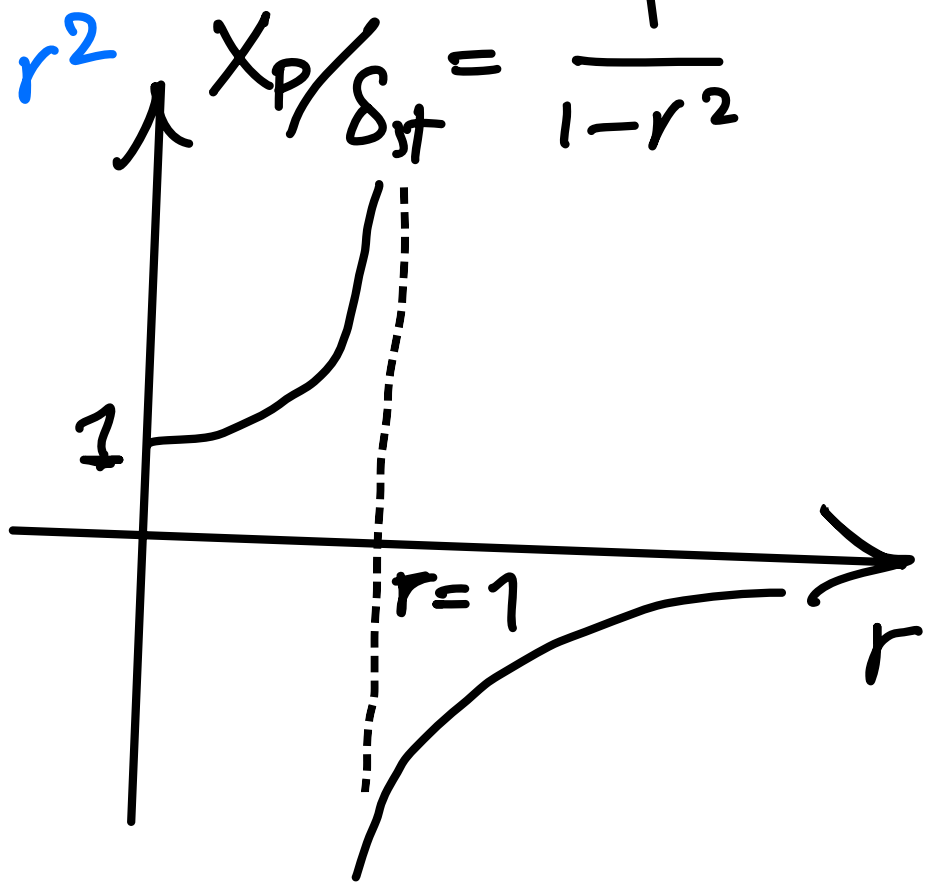
$T < T_n$

Whiteboard notes

$$|X_p(t)| = \frac{F_0}{k - m\omega^2} = \frac{F_0/k}{(k - m\omega^2)/k} = \frac{\delta_{st}}{1 - \frac{m}{k}\omega^2}$$

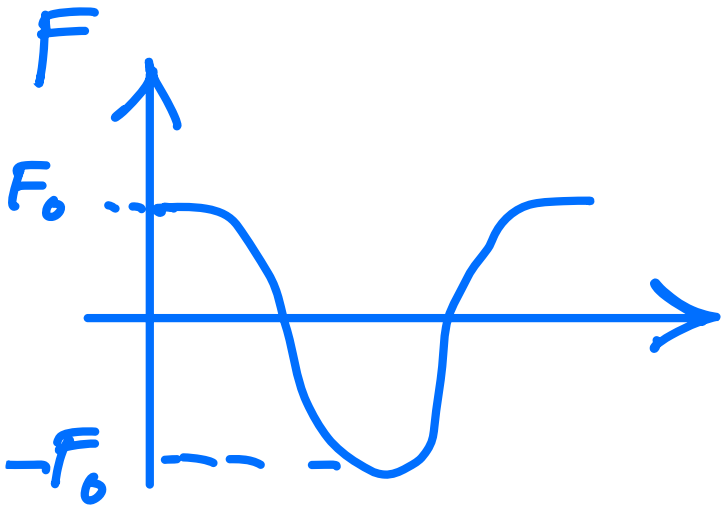
$$|X_p(t)| = \frac{\delta_{st}}{1 - \left(\frac{\omega}{\omega_n}\right)^2} = \frac{\delta_{st}}{1 - r^2}$$

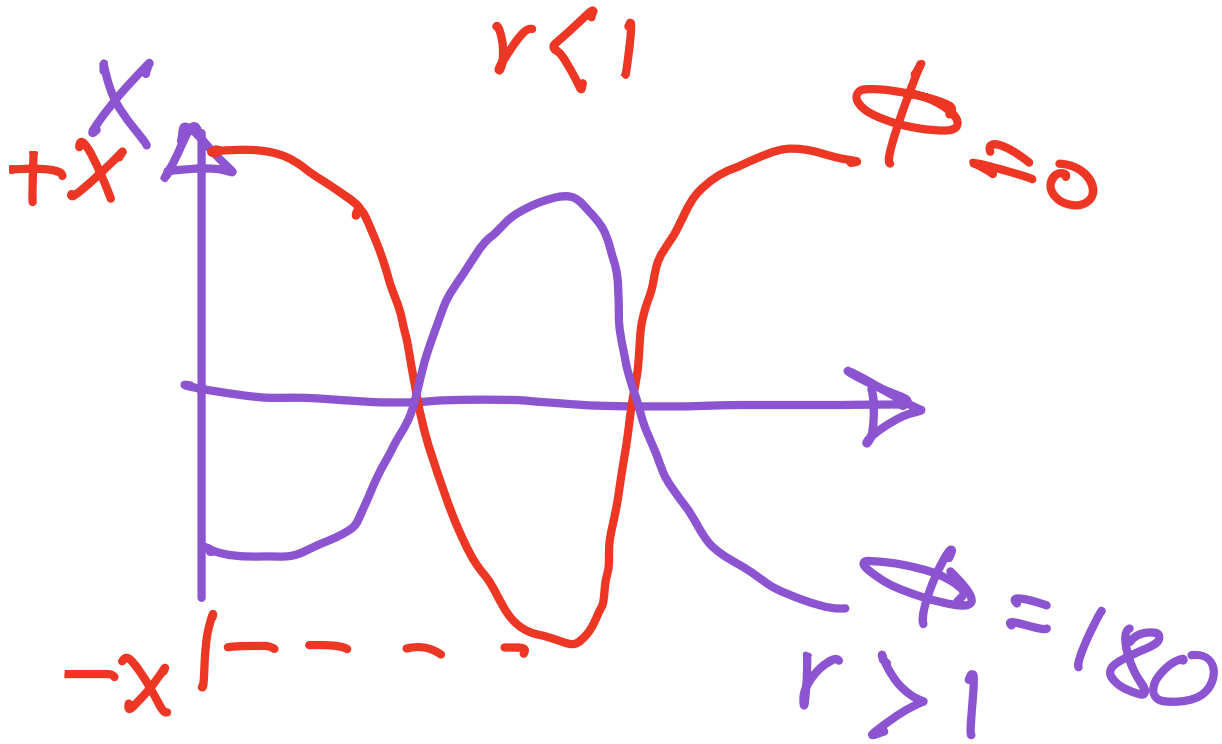
r : frequency ration



Whiteboard notes

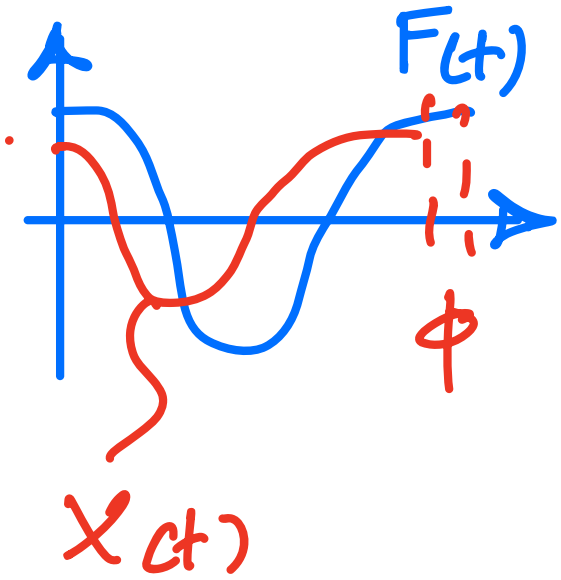
$$\frac{X_p}{\delta_{st}} = \frac{X_p}{F_0/k} = \frac{kX_p}{F_0}$$





5.3 Viscously damped response

Amplitude and phase of the forced motion



$$x_p(t) = X\cos(\omega t - \phi)$$

$$X = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (c\omega)^2}}$$

$$\phi = \text{atan2}(c\omega,\ k - m\omega^2)$$

$c = 0$

$X = \dfrac{F_0}{k - m\omega^2}$ ✓

$\phi = \tan^{-1}\dfrac{0}{k - m\omega^2}$

Why must the phase quadrant be checked?

Whiteboard notes

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos \omega t$$

$$x_h = A e^{-\zeta \omega_n t} \cos(\omega_d t - \phi) = B_1 e^{-\zeta \omega_n t} \sin \omega_d t + B_2 e^{-\zeta \omega_n t} \cos \omega_d t$$

$$\omega_d = \omega_n \sqrt{1 - \zeta^2} \qquad 0 < \zeta < 1$$

$$x_p = X \cos \omega t \qquad \dot{x}_p = -X \omega \sin \omega t \qquad \ddot{x}_p = -X \omega^2 \cos \omega t$$

$$-mX\omega^2 \cos \omega t - cX\omega \sin \omega t + kX \cos \omega t = F_0 \cos \omega t$$

Not possible!

Whiteboard notes

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos\omega t$$

$$x_p = X_1 \cos\omega t + X_2 \sin\omega t \overset{\text{or}}{=} X\cos(\omega t - \phi)$$

$$\dot{x}_p = -X\omega \sin(\omega t - \phi) \qquad \ddot{x}_p = -X\omega^2 \cos(\omega t - \phi)$$

$$-mX\omega^2\cos(\omega t-\phi) - cX\omega\sin(\omega t-\phi) + kX\cos(\omega t-\phi) = F_0\cos\omega t$$

$$\cos(\omega t - \phi) = \cos\omega t\cos\phi + \sin\omega t\sin\phi$$

$$\sin(\omega t - \phi) = \sin\omega t\cos\phi - \sin\phi\cos\omega t$$

Whiteboard notes

$$-mX\omega^2\left[\cos\omega t\cos\phi+\sin\omega t\sin\phi\right]$$

$$-cX\omega\left[\sin\omega t\cos\phi-\sin\phi\cos\omega t\right]$$

$$+kX\left[\cos\omega t\cos\phi+\sin\omega t\sin\phi\right]=F_0\cos\omega t$$

$\sin\omega t$: $-mX\omega^2\sin\phi-cX\omega\cos\phi+kX\sin\phi=0$

$\cos\omega t$: $-mX\omega^2\cos\phi+cX\omega\sin\phi+kX\cos\phi=F_0$

Whiteboard notes

$$X\left[K - m\omega^2\right]\sin\phi - cX\omega\cos\phi = 0$$

$$X\left[K - m\omega^2\right]\cos\phi + cX\omega\sin\phi = F_0$$

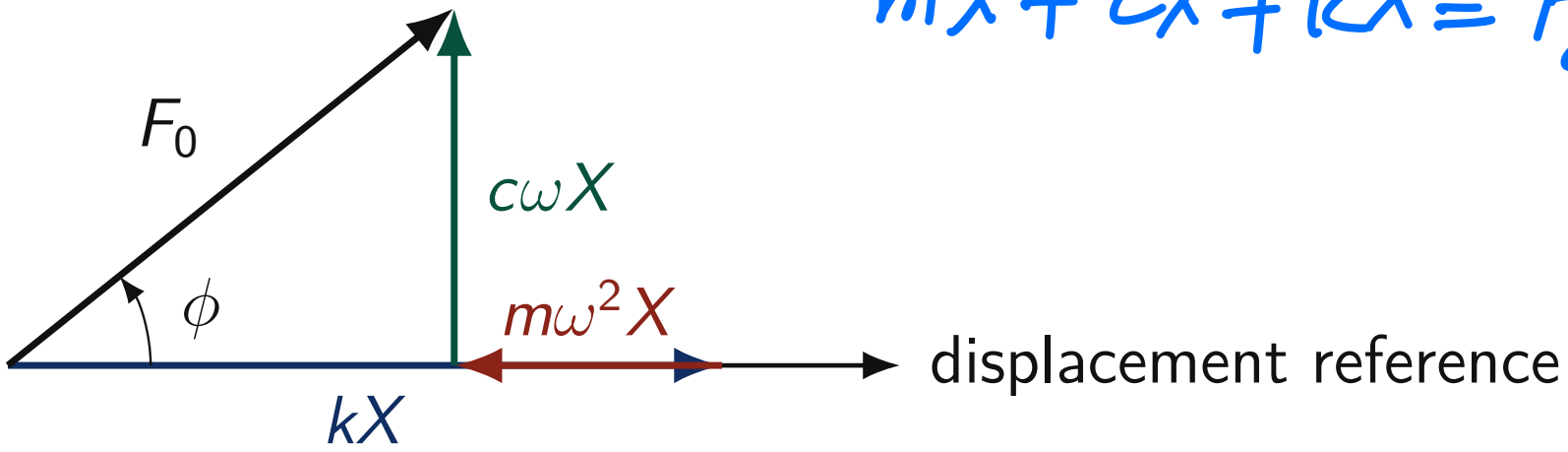
$$X = \frac{F_0}{\sqrt{\left(K - m\omega^2\right)^2 + c^2\omega^2}}$$

$$\phi = tg^{-1}\frac{c\omega}{K - m\omega^2}$$

5.3 Viscously damped response

A force-vector balance explains phase

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos \omega t$$



The applied force leads displacement by ϕ.

Whiteboard notes

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos \omega t$$

$$x(t) = A e^{-\zeta \omega_n t} \cos(\omega_d t - \alpha) + X \cos(\omega t - \phi)$$

$$X = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + c^2\omega^2}} \qquad \phi = tg^{-1} \frac{c\omega}{k - m\omega^2}$$

Whiteboard notes

$$X = \frac{F_0/k}{\sqrt{(k-m\omega^2)^2/k^2 + c^2\omega^2/k^2}} = \frac{\delta_{st}}{\sqrt{\left(1-\left(\frac{\omega}{\omega_n}\right)^2\right)^2 + \left(2\zeta\frac{\omega}{\omega_n}\right)^2}}$$

$$\frac{c^2\omega^2}{k^2} = \frac{\zeta^2 c_c^2 \omega^2}{k^2} = \frac{\zeta^2 4km\omega^2}{k^2} = \frac{\zeta^2 4m\omega^2}{k}$$

$$c_c = 2\sqrt{km} \qquad = \left(2\zeta\frac{\omega}{\omega_n}\right)^2$$

Whiteboard notes

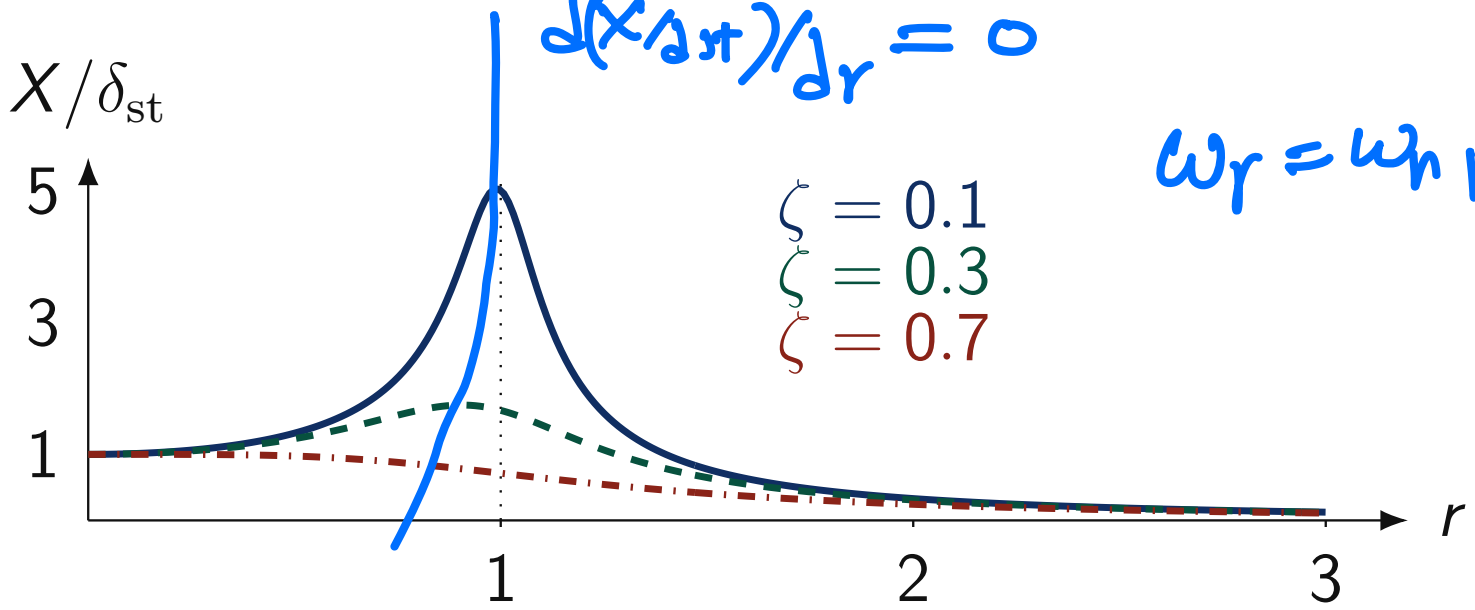
$$r = \frac{\omega}{\omega_n} \qquad X/\delta_{st} = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}$$

$$\phi = tg^{-1}\frac{c\omega}{k - m\omega^2} = tg^{-1}\frac{2\zeta r}{1-r^2}$$

5.4 Frequency-response curves

Dynamic magnification

$$r = \frac{\omega}{\omega_n}, \qquad \delta_{\text{st}} = \frac{F_0}{k}, \qquad \frac{X}{\delta_{\text{st}}} = \frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}$$

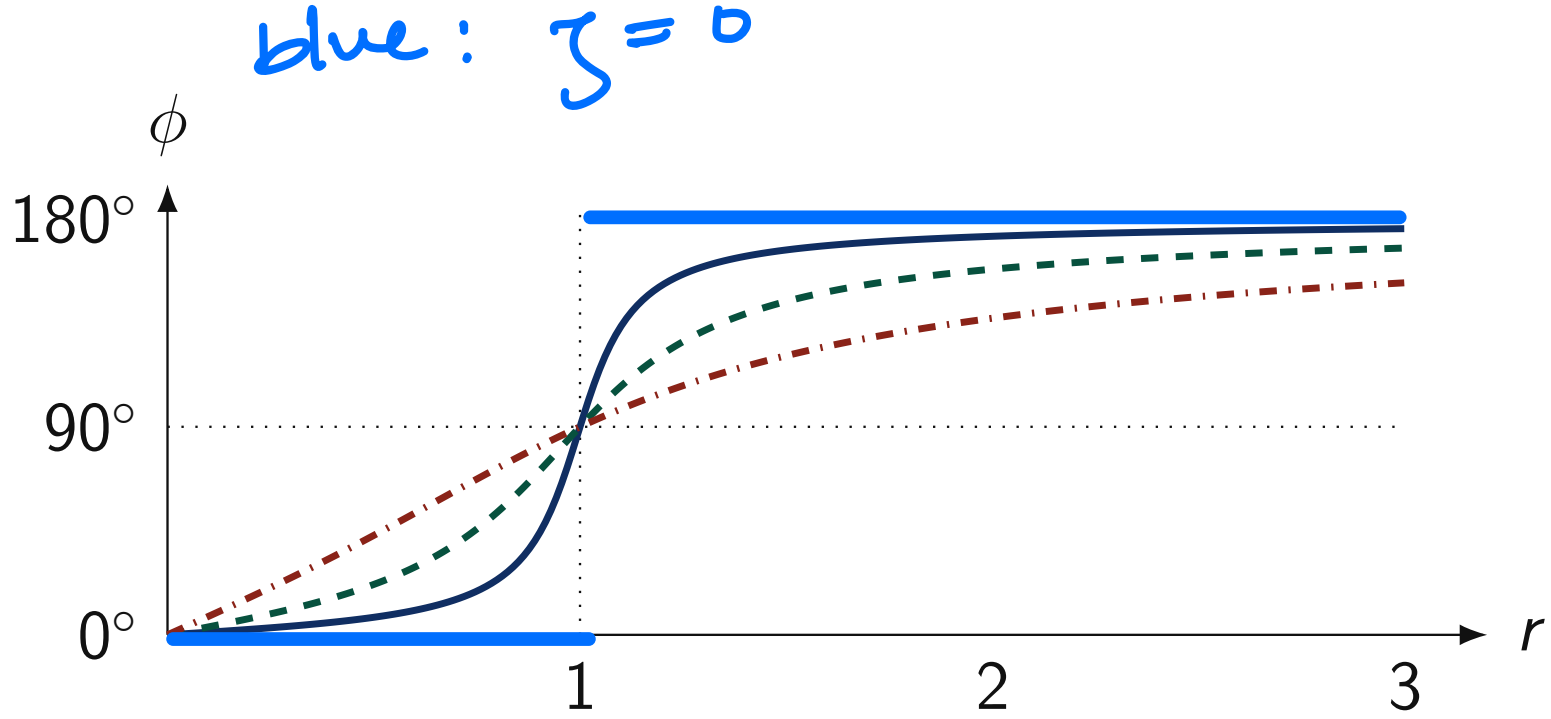


$d(X/\delta_{st})/dr = 0$

$\omega_r = \omega_n\sqrt{1-2\zeta^2}$

5.4 Frequency-response curves

Phase lag changes through resonance



$\zeta = 0.1$ (solid) 0.3 (dashed) 0.7 (dash-dot)

$r \ll 1$: in phase $r = 1$: 90° $r \gg 1$: opposite phase

5.5 Resonance and damping

Peak displacement is not exactly at ω_n

$$\omega_{\text{peak}} = \omega_n\sqrt{1-2\zeta^2}, \qquad 0 < \zeta < \frac{1}{\sqrt{2}}$$

$$\left(\frac{X}{\delta_{\text{st}}}\right)_{\text{max}} = \frac{1}{2\zeta\sqrt{1-\zeta^2}} \approx \frac{1}{2\zeta}$$

$\zeta \ll 1$

At $\omega = \omega_n$: $X/\delta_{\text{st}} = 1/(2\zeta)$.

Distinguish ω_d, ω_n, and ω_{peak}.

5.5 Resonance and damping

Complex notation gives the same result

$$F(t) = \Re\{F_0 e^{i\omega t}\}, \qquad x_p(t) = \Re\{\hat{X} e^{i\omega t}\}$$

$$(k - m\omega^2 + ic\omega)\hat{X} = F_0$$

$$\hat{X} = \frac{F_0}{k - m\omega^2 + ic\omega} = X e^{-i\phi}$$

Take the real part to recover the physical displacement.

Whiteboard notes

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos\omega t = \mathrm{Re}\left[f_0 e^{i\omega t}\right]$$

$$m\ddot{x} + c\dot{x} + kx = F_0 e^{i\omega t}$$

$$x(t) = Xe^{i\omega t} \qquad \dot{x} = Xi\omega e^{i\omega t} \qquad \ddot{x} = -X\omega^2 e^{i\omega t}$$

$$-mX\omega^2 e^{i\omega t} + Xic\omega e^{i\omega t} + kXe^{i\omega t} = F_0 e^{i\omega t}$$

$$X = \frac{F_0}{k - m\omega^2 + ic\omega}$$

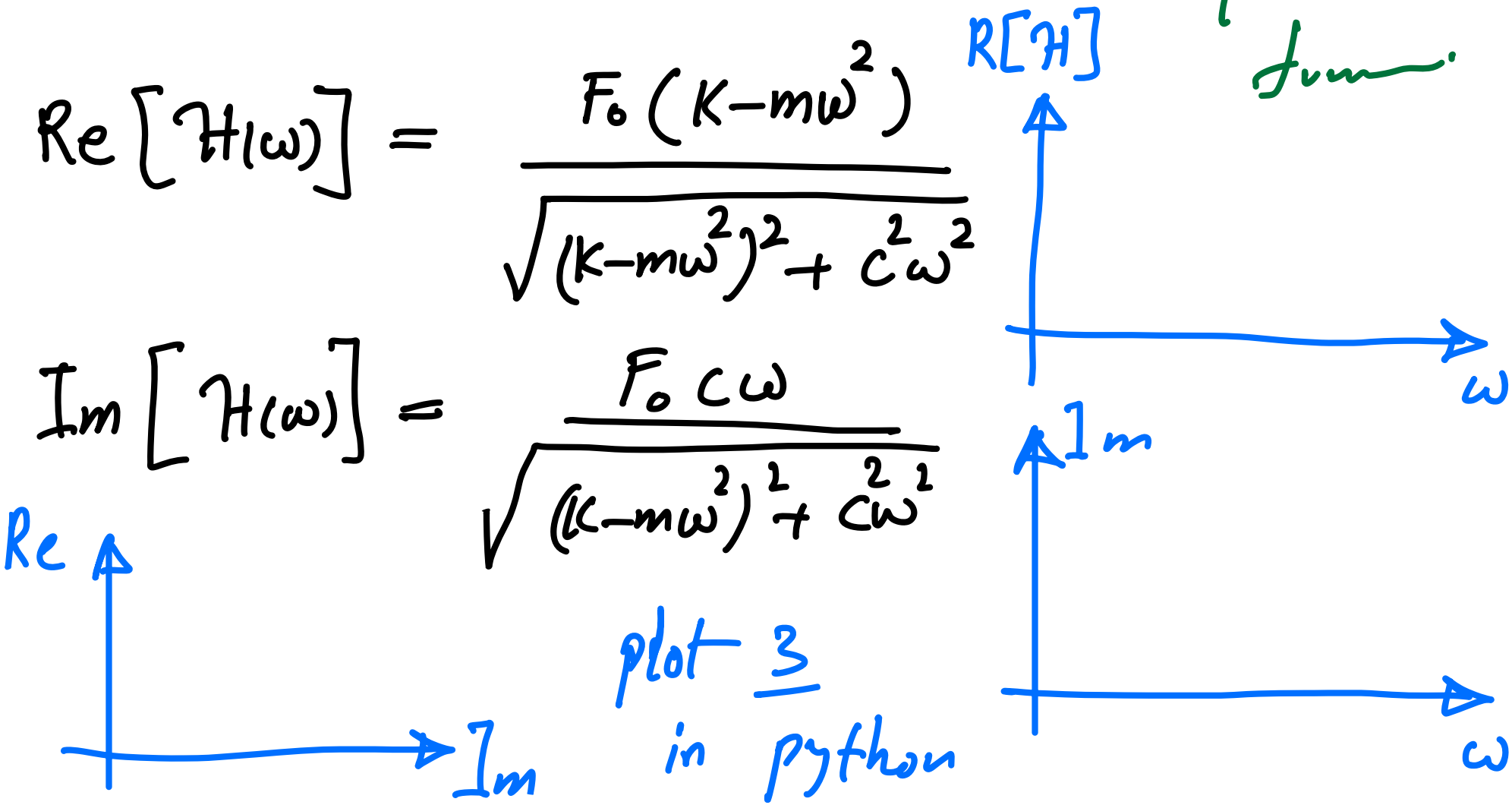
Whiteboard notes

$$X = \frac{F_0}{K - m\omega^2 + i c\omega} = \mathcal{H}(\omega)$$

frequency response fun.

$$\mathrm{Re}\left[\mathcal{H}(\omega)\right] = \frac{F_0 (K - m\omega^2)}{\sqrt{(K - m\omega^2)^2 + c^2\omega^2}}$$

$$\mathrm{Im}\left[\mathcal{H}(\omega)\right] = \frac{F_0 c\omega}{\sqrt{(K - m\omega^2)^2 + c^2\omega^2}}$$



plot 3 in python

Whiteboard notes

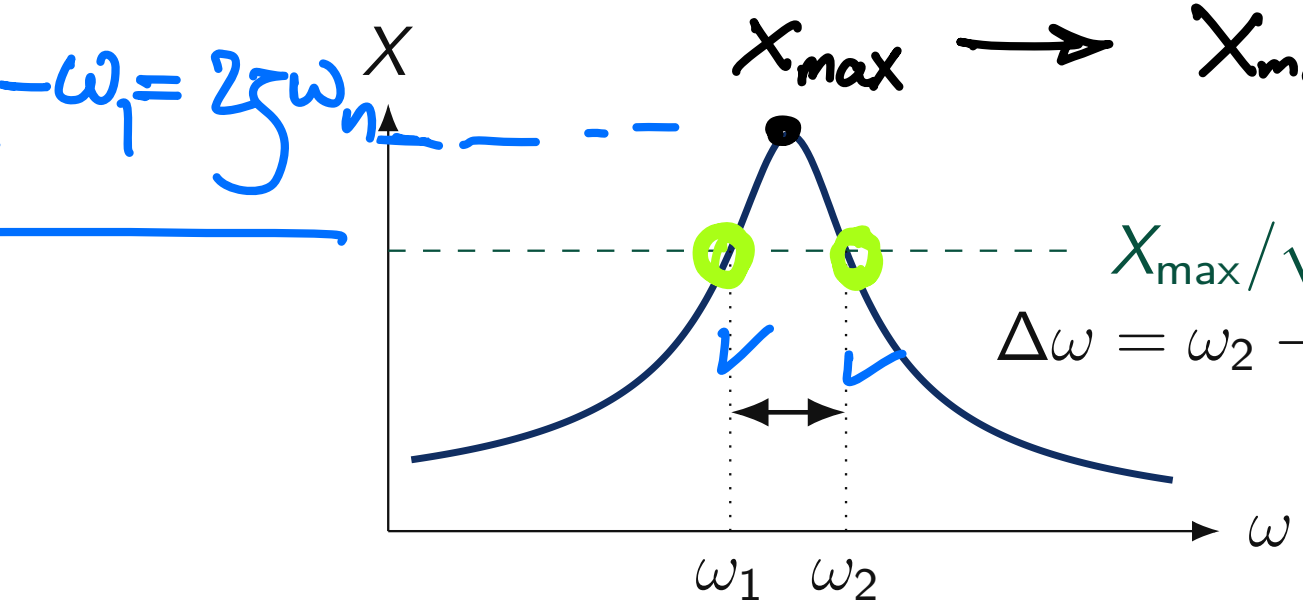
$$H(\omega) = \frac{F_0}{k - m\omega^2 + i c\omega}$$

$$|H(\omega)| = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + c^2\omega^2}}$$

$$\phi = 0 - \tan^{-1}\frac{c\omega}{k - m\omega^2}$$

5.5 Resonance and damping

Quality factor and bandwidth: estimate ζ



$$Q \approx \frac{1}{2\zeta} \approx \frac{\omega_n}{\Delta\omega}, \qquad \boxed{\zeta \approx \frac{\omega_2 - \omega_1}{2\omega_n}}$$

Light damping; constant force amplitude. Use $\omega_n \approx \omega_{\text{peak}}$.

5.6 Worked application and checks

Book example: total response

$$m = 10\ \mathrm{kg}, \quad c = 20\ \mathrm{N\,s/m}, \quad k = 4000\ \mathrm{N/m}$$

$$F(t) = 100\cos(10t)\ \mathrm{N}, \quad x_0 = 0.01\ \mathrm{m}, \quad v_0 = 0$$

Find the steady response, then enforce both initial conditions.

Rao, Example 3.3(a)

Whiteboard notes

$$m\ddot{x} + c\dot{x} + kx = F_0 \cos \omega t$$

$$10\ddot{x} + 20\dot{x} + 4000x = 100 \cos 10t$$

$$\ddot{x} + 2\dot{x} + 400x = 10 \cos 10t$$

$$2\zeta\omega_n = 2 \qquad \zeta = 5\% = 0.05$$

$$\omega_n^2 = 400 \rightarrow \omega_n = 20 \text{ rad/s}$$

underdamped.

Whiteboard notes

$$x_h = A e^{-\zeta \omega_n t} \cos(\omega_d t - \alpha)$$

$\zeta = 0.05$

$\omega_n = 20 \text{ rad/s}$

$$\omega_d = \omega_n \sqrt{1-\zeta^2}$$

$$x_p = X \cos(\omega t - \phi)$$

$$X = \frac{100}{\sqrt{(4000 - 10(10)^2)^2 + 20^2 \cdot 10^2}} \checkmark$$

$$\phi = tg^{-1} \frac{20 \cdot 10}{4000 - 10 \cdot 10^2} \checkmark$$

Whiteboard notes

$$x_{tot} = A e^{-\zeta \omega_n t} \cos(\omega_d t - \alpha) + X \cos(\omega t - \phi)$$

$A = ?$

$\alpha = ?$

$x(0) = 0.01 \text{ m}$

$\dot{x}(0) = 0$

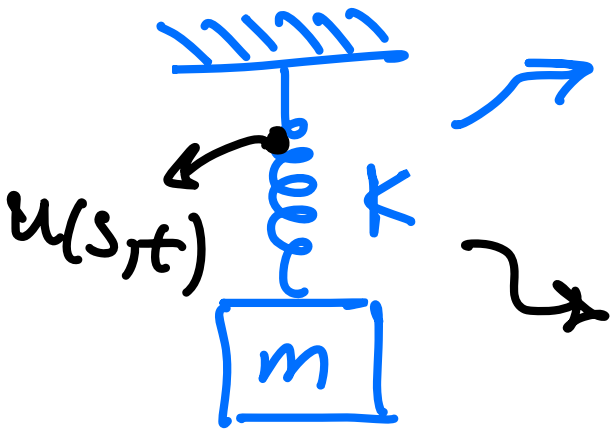
5.6 Worked application and checks

Before accepting the result

- Do $x(0)$ and $\dot{x}(0)$ match the prescribed values?
- Is the phase in the correct quadrant?
- Does the low-frequency limit give F_0/k?
- Is the resonant case treated separately when $c = 0$?

Whiteboard notes

Recitation 3, 2G



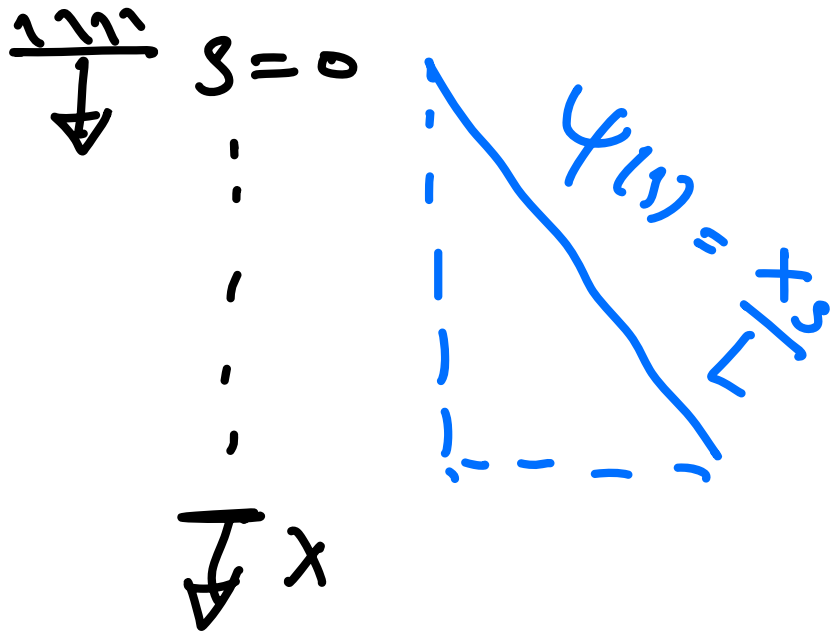
→ motion

↳ 1 DOF

↳ has mass

∞ DOF

Continuous.



$u(s,t)$: displacement

for $s = 0$ to $s = x$

$u(s,t) = \psi(s) q(t)$

$\psi(1) =$ linear

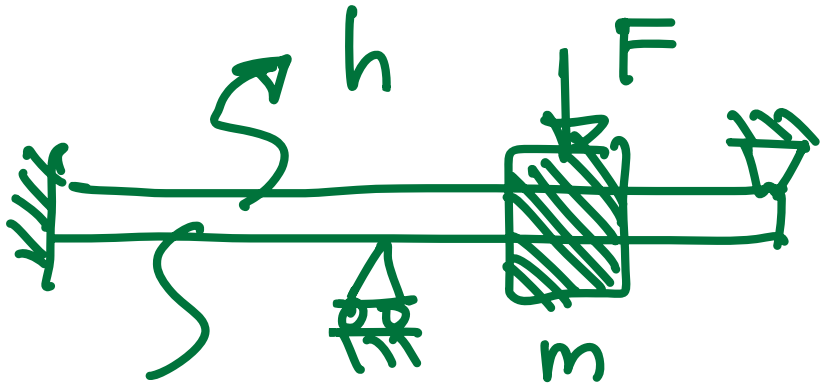
Whiteboard notes

$$T = \int dT = \int \frac{1}{2} v^2 dm = \int \frac{1}{2} \dot{u}^2 dm$$

$$= \frac{1}{2} \int_0^L \psi^2(s) \dot{q}^2(t) \rho \, dl = \frac{1}{2} m^s_{eq} \dot{q}^2$$

$$M_{eq} = m + m^s_{eq}$$

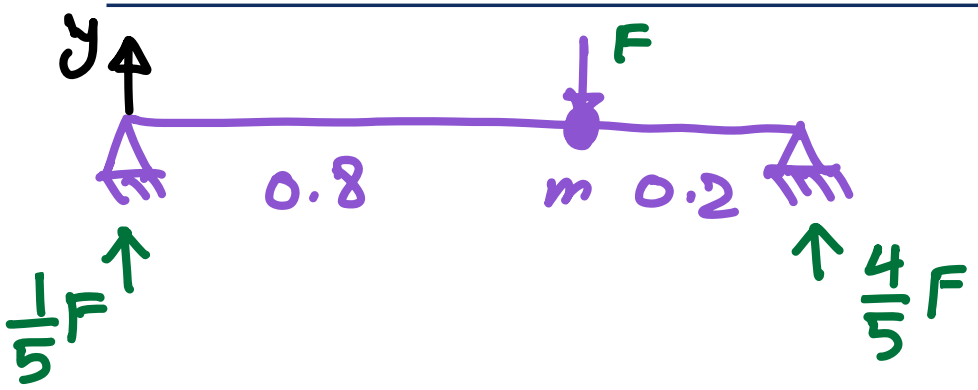
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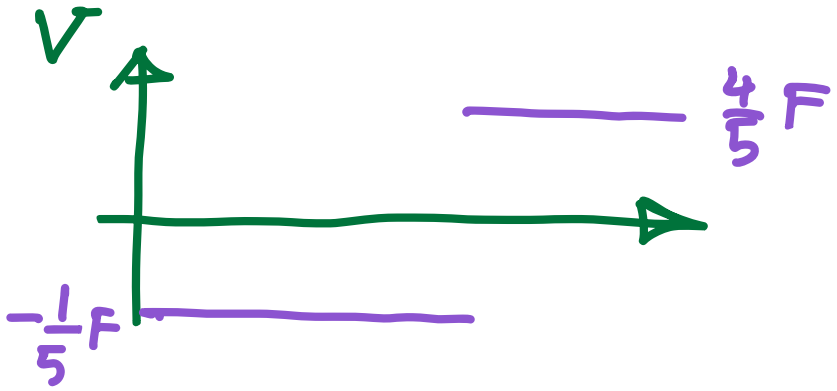


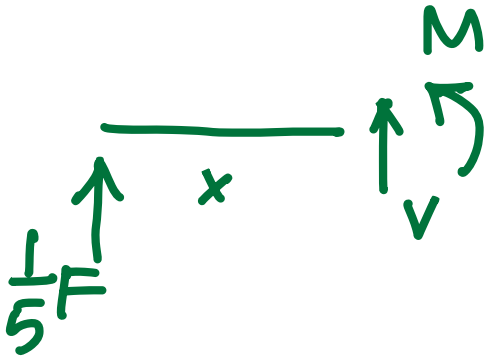
$\equiv \quad m_{eq}\ddot{x} + c_{eq}\dot{x} + k_{eq}x = F$

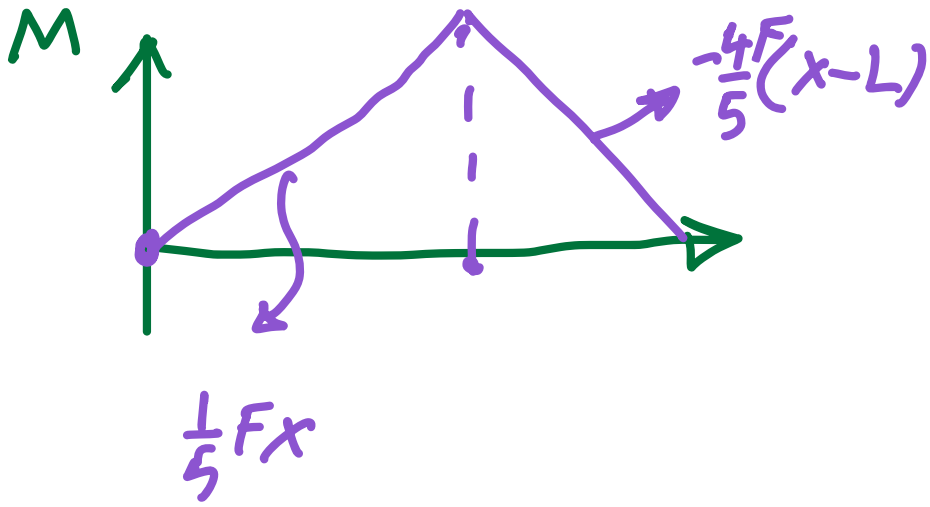
$\rho E I L$

Whiteboard notes









$M(x) = EI\,y''(x)$

$0 < x < 0.8$

$\frac{1}{5}Fx = EIy''$

$EIy' = \frac{1}{10}Fx^2 + C_1$

$EIy = \frac{1}{30}Fx^3 + C_1x + C_2$

$0.8 < x < 1$

$$EI y' = -\frac{4}{5} F(x-L)$$

$$EI y' = -\frac{4}{5} F\left(\frac{x^2}{2} - Lx\right) + C_3$$

$$EI y = -\frac{4}{5} F\left(\frac{x^3}{3} - \frac{Lx^2}{2}\right) + C_3 x + C_4$$

$y(L) = 0 \rightarrow$

$y_1(0.8) = y_2(0.8) \rightarrow$

$y_1'(0.8) = y_2'(0.8) \rightarrow$

$\left\{ \begin{array}{l} C_4 \checkmark \\ C_3 \checkmark \\ C_4 \checkmark \end{array} \right.$

$u(x,t) = \psi(x) \cdot q(t)$

$\psi(s) \simeq y(x) \rightarrow$ $\int$ $\int$